

Stopping Revenue Leakage

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The Problem

Acquiring a new telecom customer costs 5 to 25 times more than keeping an existing one. Customer churn quietly destroys monthly recurring revenue before sales teams even notice a problem.

The Solution

The Telco Churn Engine—an intelligent prediction system built specifically on customer behavior patterns to identify churn risk weeks before a customer cancels.

Why It Matters

Shifts your customer success strategy from reactive damage control to proactive, high-ROI retention campaigns.

Key Metric

5-25x

Higher cost for new acquisition vs. retention.

"Protect the customers you've already paid to acquire."

4-in-1 Engine Technology

Multi-Model Tournament

Instead of relying on a single algorithm, our architecture benchmarks 4 industry-leading gradient boosting engines (AdaBoost, XGBoost, LightGBM, and CatBoost) simultaneously.

Automated Fine-Tuning

Powered by Optuna hyperparameter optimization, the system continuously calibrates every engine to find the highest possible predictive accuracy.

Leaderboard Transparency

An automated ranking system guarantees that only the top-performing engine powers your daily business decisions.

AdaBoost

XGBoost

LightGBM

CatBoost

Battle-Tested Reliability & ROI Protection

Stress-Tested Realism

Validated using **5-Fold Stratified Cross-Validation** to ensure the AI doesn't just memorize past data, but performs reliably on future, unseen customers.

Precision Over Fluff

Evaluated on F1-Score and ROC-AUC rather than misleading raw accuracy scores. This ensures you don't offer unnecessary discounts to loyal customers.

Budget Protection

Maximizes retention ROI by directing your promotional and marketing budget strictly toward true churn risks.

Turnkey Business Dashboard (Zero Technical Friction)

No-Code Streamlit App

Deployed as an intuitive, interactive business dashboard designed for marketing, sales, and account managers.

Instant Risk Scoring

Customer service representatives can input account details into the Prediction Tab to receive an instant churn probability score and offer tailored retention deals on the spot.

Performance Visibility

The Leaderboard Tab gives executives full visibility into system metrics, confusion matrices, and model health.

Actionable Intelligence & Rapid Deployment

Root-Cause Analysis

Automated Feature Importance charts expose why customers leave (e.g., contract types or pricing tiers), giving leadership clear policy recommendations.

Frictionless Integration

Built on modern, lightweight Python infrastructure using `uv` environment isolation for effortless deployment into your existing CRM workflows.

The Bottom Line

- Protect recurring revenue
- Optimize retention spending
- Make data-backed decisions from day one